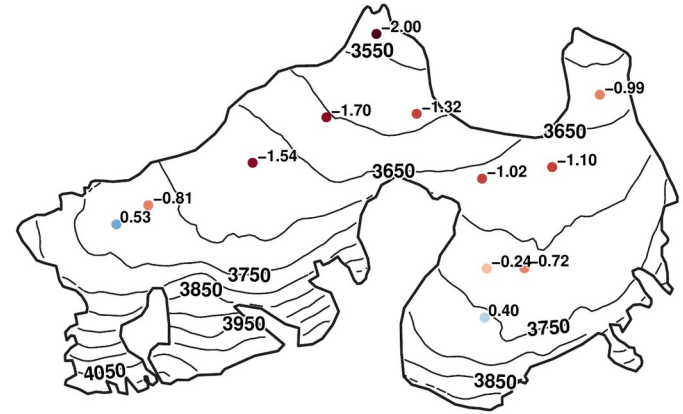


Modeling glacier mass balance from point measurements

Enrico Mattea

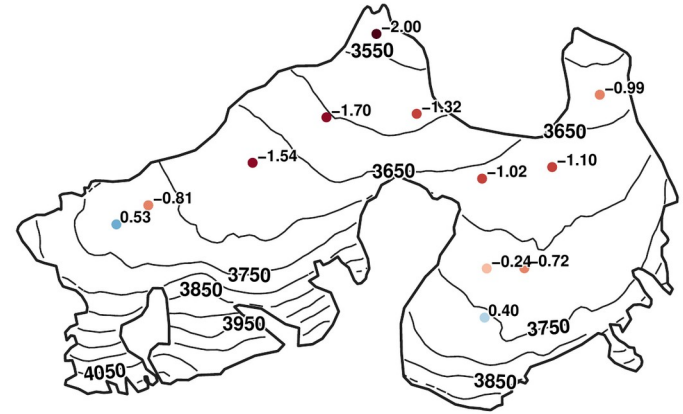
Introduction

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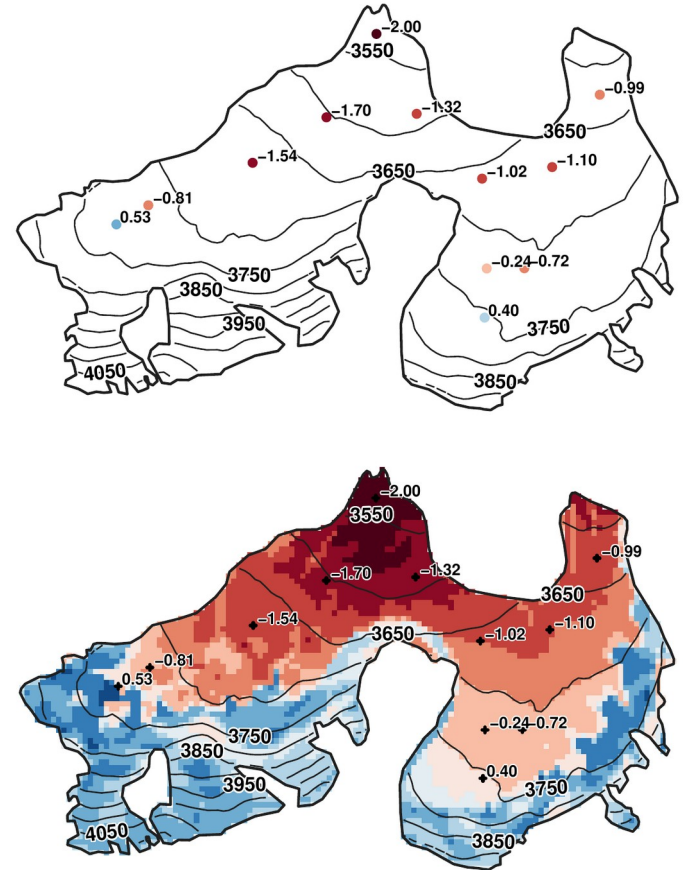


Introduction

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Mass balance model



Mass balance model

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- **Accumulation** = Snowfall (rain/snow and total amount depend on altitude and topography)
- **Ablation** = Melt (depends on air temperature → altitude, but also on solar radiation → topography)
- **Avalanches** → Snow redistribution (also outside the glacier)



Mass balance model

- The model simulates **snowfall, melt and avalanches**

Mass balance model

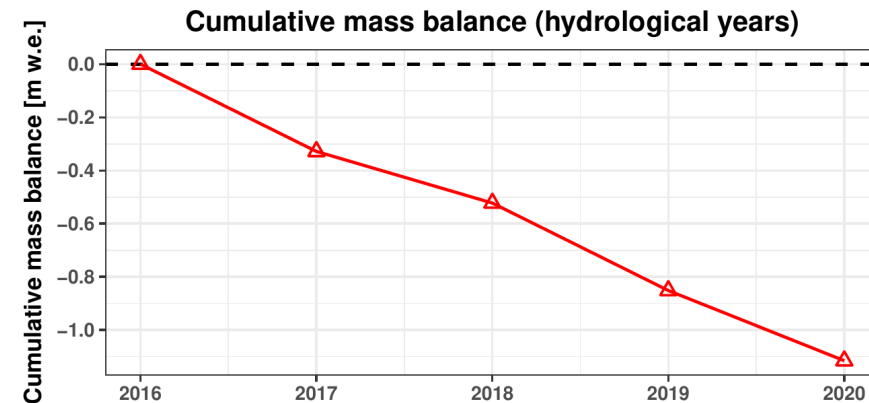
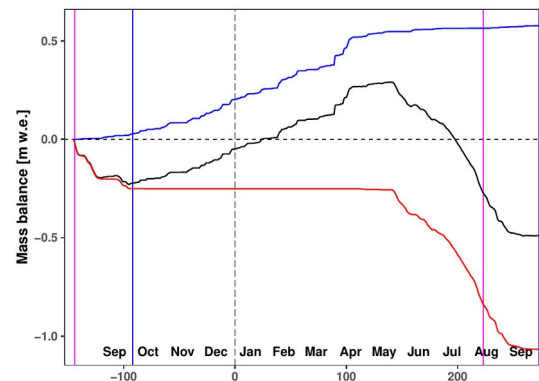
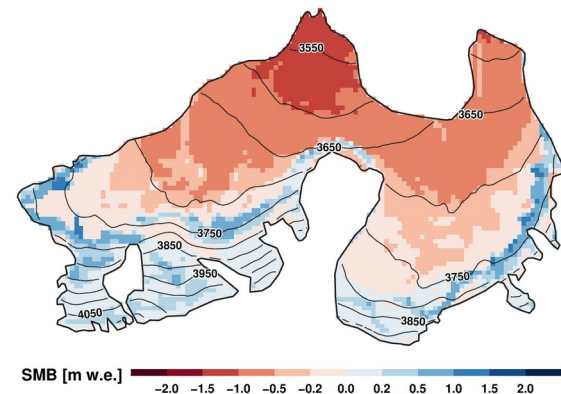
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Mass balance model

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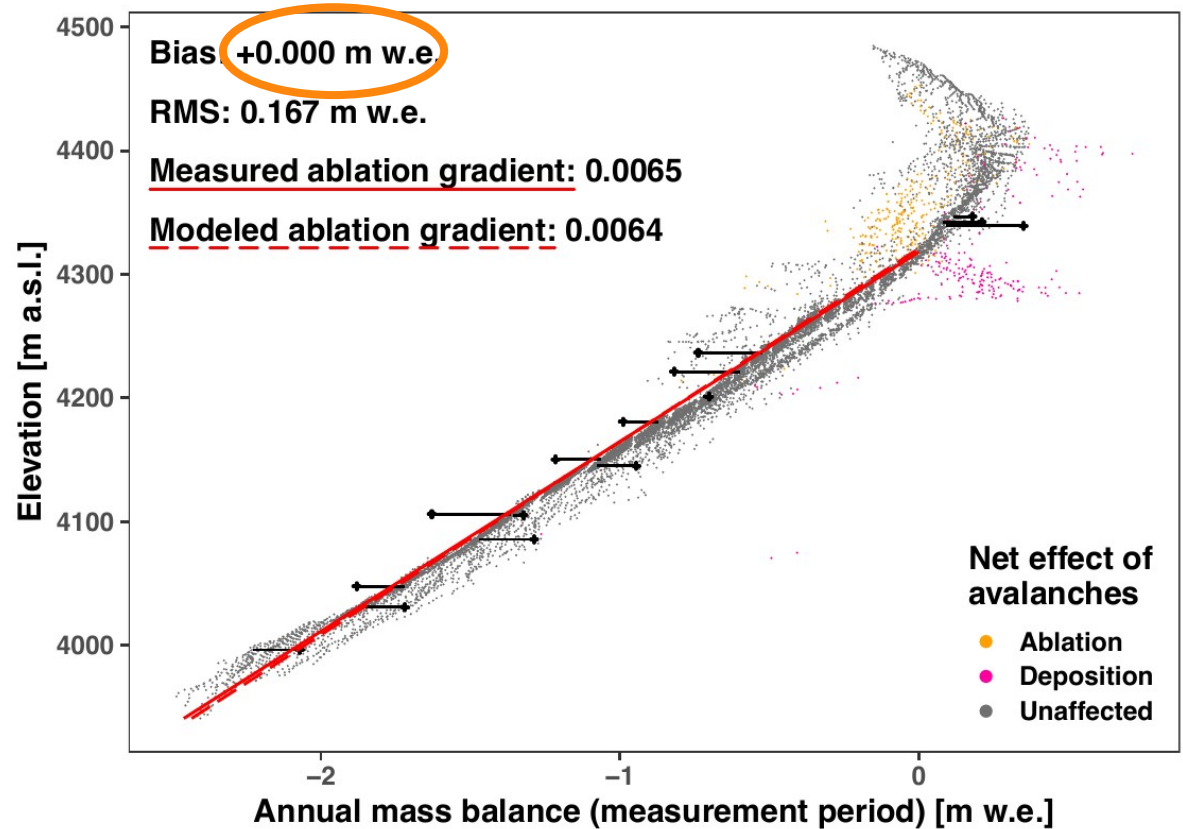
Mass balance model

- The model simulates **snowfall, melt and avalanches**
- On a **grid of cells** which cover the glacier
- At **daily resolution**
- **Output:** mass balance maps, time series, comparison with the measured points



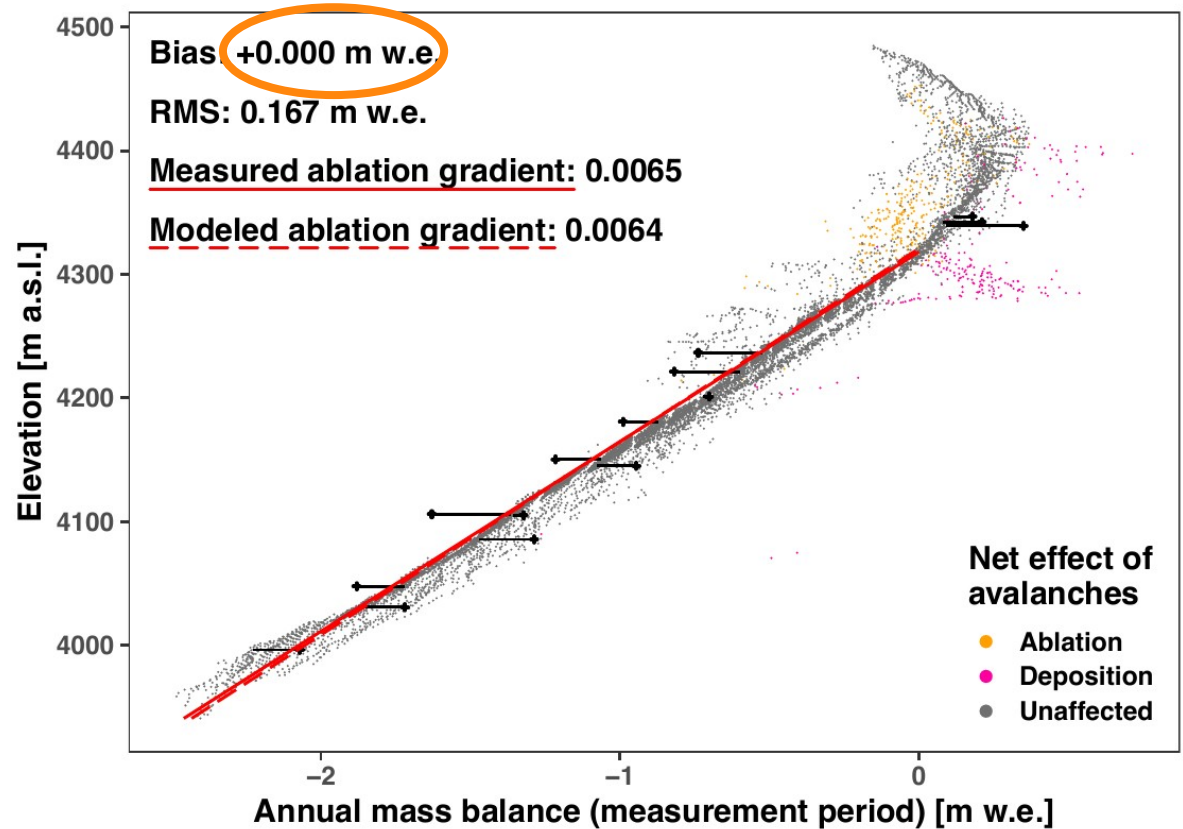
Mass balance model

The simulation **optimizes**
the melt parameters
automatically, until there
is **no bias** compared to
the measured points
(mean modeled mass
balance = mean
measured mass balance)



Mass balance model

If measurements in winter are available, the model first **optimizes the factor of precipitation correction**, until there is no bias over winter (when there is generally little to no melt)



How to use the model

- The model uses the  script language: free, open-source

How to use the model

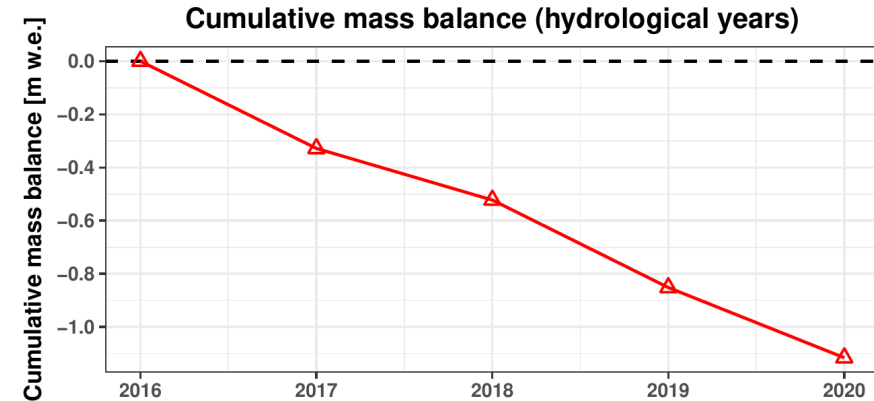
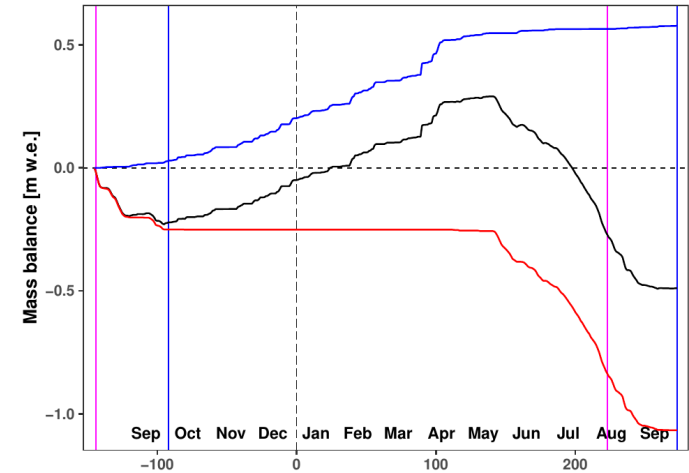
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How to use the model

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- You have to **install some programs** to use the model: R, RTools, RStudio
- Installation is described in the **model tutorial 1**

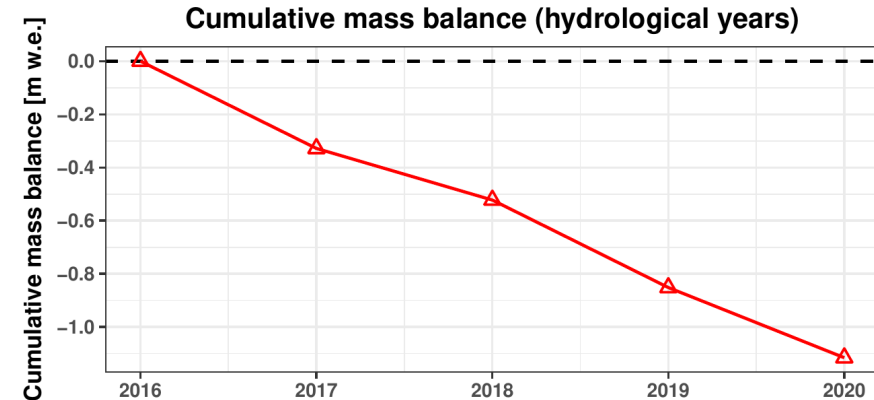
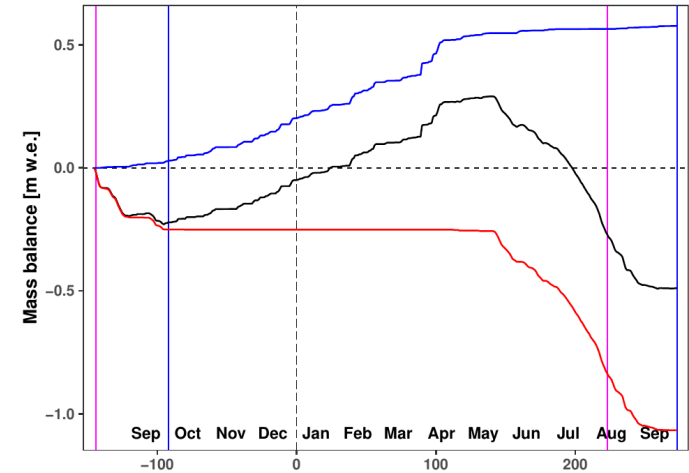
How to use the model

- The model can simulate one **single year** or **several years**



How to use the model

- The model can simulate one **single year** or **several years**
- For several years, the model also shows the **cumulative mass balance**



How to use the model

- Prepare the **input files**

How to use the model

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- Setup the model **parameters**

How to use the model

- Prepare the **input files**
- Setup the model **parameters**
- **Run** the model!

Input files

- Digital elevation model (DEM)
- Surface type (ice, firn, debris)
- Daily solar radiation (computed from the DEM)

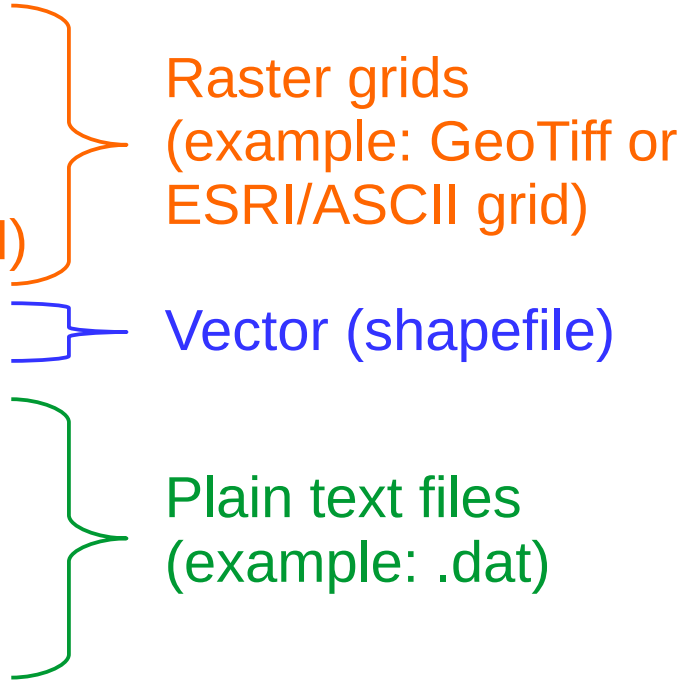


Raster grids
(example: GeoTiff or
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Input files

- Digital elevation model (DEM)
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-
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Input files

- Digital elevation model (DEM)
 - Surface type (ice, firn, debris)
 - Daily solar radiation (computed from the DEM)
 - Glacier outline
 - Meteorological series (daily temperature and precipitation)
 - Mass balance measurements (stakes, snow pits, snow probes)
- 
- The diagram uses colored brackets to group the input files into three categories:
- Raster grids** (example: GeoTiff or ESRI/ASCII grid): This category includes the first three items in the list: Digital elevation model (DEM), Surface type (ice, firn, debris), and Daily solar radiation (computed from the DEM). It is indicated by an orange bracket.
 - Vector (shapefile)**: This category includes the fourth item: Glacier outline. It is indicated by a blue bracket.
 - Plain text files** (example: .dat): This category includes the last three items: Meteorological series (daily temperature and precipitation), Mass balance measurements (stakes, snow pits, snow probes), and an unlabeled item. It is indicated by a green bracket.

Input files

- To prepare the input files: use the tool *make_input.R*

The screenshot shows a web browser window with a URL bar displaying 'http://127.0.0.1:3705'. The page contains a form for configuring glacier modeling parameters. The form includes several input fields and buttons. The 'Choose glacier name' field contains 'yakarcha'. The 'Choose model year' field contains '2020'. Below these are five buttons: 'Choose one or more input DEM files', 'Choose input glacier shapefile', 'Choose input firn shapefile (optional)', 'Choose input debris shapefile (optional)', and 'Choose reference grid file (optional)'. The 'Choose margin size around the outline, in meters:' field contains '100'. The 'Choose grid cell size in meters (leave blank for automatic cell size):' field is empty. A checkbox labeled 'Compute daily potential solar radiation (SLOW!)' is checked. Below the checkbox, there are several status messages: 'Glacier name selected.', 'Model year selected.', 'Input DEM file(s) selected.', 'Input outline file selected.', 'Input firn shapefile (optional) not yet selected.', 'Input debris shapefile (optional) not yet selected.', and 'Input reference grid file (optional) not yet selected.'. At the bottom of the form is a button labeled 'RUN!'.

http://127.0.0.1:3705 Open in Browser Publish

Choose glacier name: yakarcha

Choose model year: 2020

Choose one or more input DEM files

Choose input glacier shapefile

Choose input firn shapefile (optional)

Choose input debris shapefile (optional)

Choose reference grid file (optional)

Choose margin size around the outline, in meters:
100

Choose grid cell size in meters (leave blank for automatic cell size):

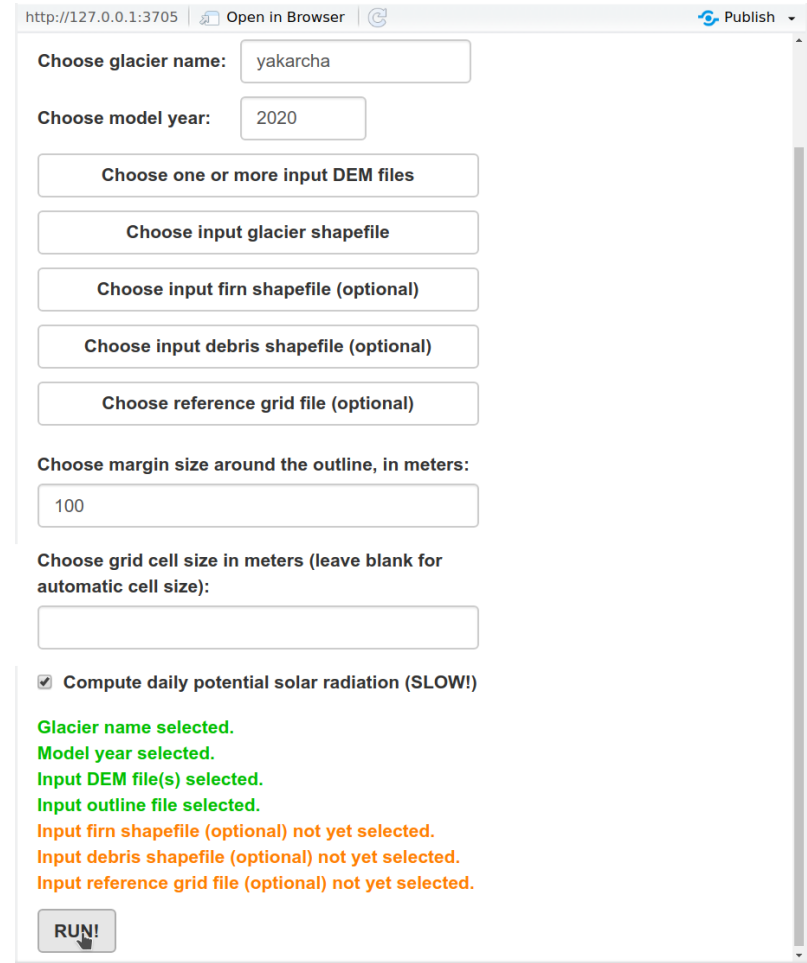
☒ Compute daily potential solar radiation (SLOW!)

Glacier name selected.
Model year selected.
Input DEM file(s) selected.
Input outline file selected.
Input firn shapefile (optional) not yet selected.
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RUN!

Input files

- To prepare the input files: use the tool *make_input.R*
- If you simulate **several years**, you can use several **DEMs**



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- Choose glacier name:** A text input field containing the value "yakarcha".
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- Choose grid cell size in meters (leave blank for automatic cell size):** A text input field.
- Compute daily potential solar radiation (SLOW!):** A checkbox that is checked.
- Status messages:** A series of green and orange text messages indicating the progress of file selection:
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- RUN! button:** A button with a mouse cursor icon.

Model parameters

- Two types of parameters: **fixed parameters** and **annual parameters**

Model parameters

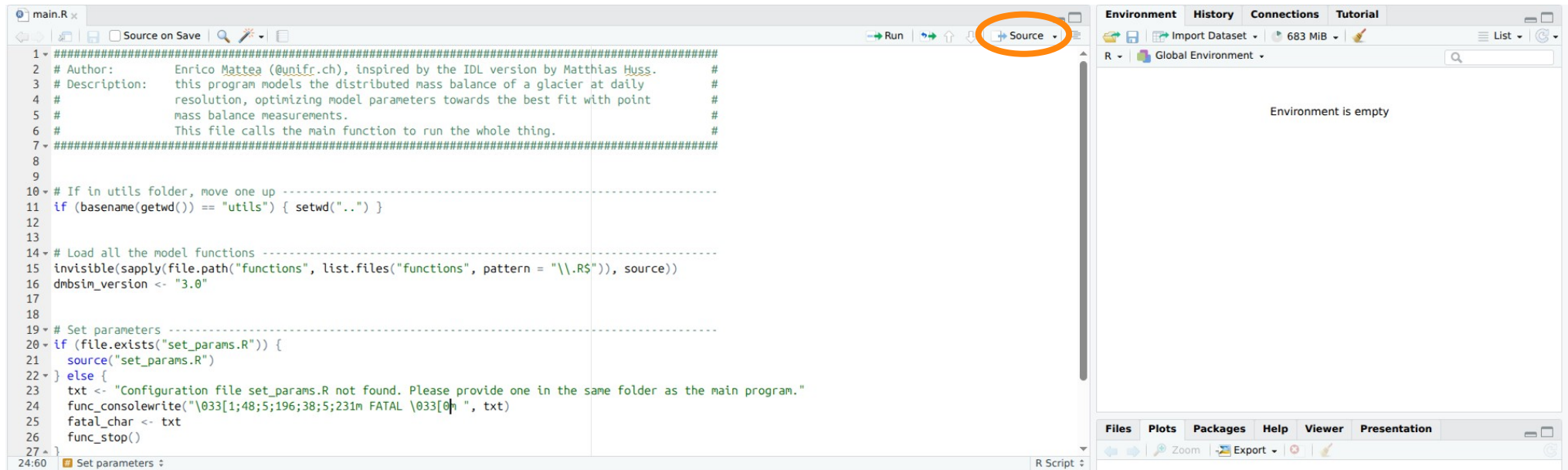
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Model parameters

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- **Fixed parameters:** glacier name, model years, AWS altitude, physical parameters (snowfall distribution, avalanches, ...)
- **Annual parameters (optional):** to use a special value of a parameter in a specific year. One file per year (example: *param_2020.dat*). Precipitation correction, melt factors, altitude bands for contour-line correction, grids of relative snow distribution

Running the model

- When input data and parameters are ready:



Running the model

- Model output:
 - **PDF** files → plots
 - **CSV** files → data of the plots
 - **TIF** files → GeoTiffs (for QGIS or ArcGIS)

Running the model

- Important output files:
 - *overview.pdf* → general plots for the full period
 - *overview_areaplot.pdf* → hydrological year mass balance
 - *annual_results/* → detailed annual results
 - *massbalance_2018.pdf* → main plots of the year
 - *daily_results/* → detailed daily results
 - *logs/* → log file (full modeling information)

Running the model

- Follow these links to the documentation to learn more:

[Tutorial 1](#) → Installation, simple run example

[Tutorial 2](#) → Run over many years

[Model description](#) → More information on the calculations

[Melt equations](#) → Detailed, graphical description of the melt formulas for different surface types

[Daily output](#) → Description of the variables in the daily output files

An aerial photograph of a massive glacier flowing through a mountain valley. The glacier is a light blue-grey color, contrasting with the dark, rocky mountain slopes on either side. The sky is a clear, pale blue.

Thank you for your attention!

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